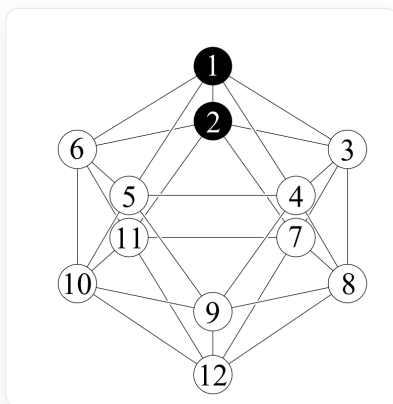


Question

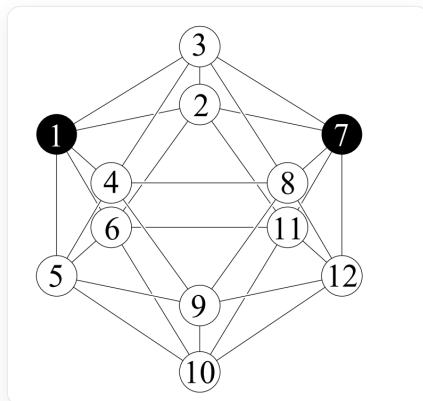
"CageWalking" Strategy is a strategy applied to metal-catalyzed functionalization of carboranes. The "cage walking" phenomenon was first observed by Hawthorne in 1971 in the nido- $C_2B_9H_{12}$ system, which means that the metal unit on the carborane cage undergoes 1,2-migration. The charge density on the C or B atoms of the carborane cage is one of the important factors controlling the migration of metal atoms. The structures of $o-C_2B_{10}H_{12}$ and $m-C_2B_{10}H_{12}$ are shown in the following figure: black represents CH_2 , white represents BH

$o-C_2B_{10}H_{12}$:



[H]C1234B567([H])C189([H])B2%10%11([H])B3%12%13([H])B54%14([H])B6%15%16([H])B87%17([H])B9%10%18([H])B%17%15%19([H])B%14%12%16([H])B%18%

$m-C_2B_{10}H_{12}$:



[H]B1234C567([H])B189([H])C2%10%11([H])B3%12%13([H])B54%14([H])B6%15%16([H])B87%17([H])B9%10%18([H])B%17%15%19([H])B%14%12%16([H])B%18%

Define the distance between two vertices on the icosahedral framework as the minimum number of edges required to connect the two vertices. Thus, the chemical environment of a boron atom can be determined based on its distance to the two carbon atoms. Use an ordered pair (a, b) to represent the chemical environment of a boron atom, where a is the smaller of the distances from the boron atom to the two carbon atoms,

b is the larger of the distances from the boron atom to the two carbon atoms, and (a, b) is called the distance pair of the boron atom.

Sort the charge densities of all B atoms in the two carboranes from high to low, respectively (boron atoms with the same distance pair are recorded only once), and map the two groups of sorted distance pairs to points in the Cartesian coordinate system, respectively. Connect the points in order to obtain two sets of broken lines. The total length of the broken line obtained by $o-C_2B_{10}H_{12}$ is denoted as x , and the total length of the broken line obtained by $m-C_2B_{10}H_{12}$ is denoted as y . Then the value of $\frac{x+y}{x-y}$ is:

A. All other options are incorrect

B. 0

C. 5.4

D. 10.8

E. 15.5

F. 20.9

G. -5.4

H. -10.8

I. -15.5

J. -20.9

Answer

Correct Answer: **I**

Detailed Explanation

C's electronegativity is greater than B's, mainly considering C's inductive electron-withdrawing effect.

CHECKPOINT

1 PTS

Considering C's inductive electron-withdrawing effect

Therefore, the farther away from the C atom, the greater the charge density of the B atom;

CHECKPOINT

1 PTS

The farther away from the C atom, the greater the charge density of the B atom

Based on this, the charge density of the B atoms in $o-C_2B_{10}H_{12}$ can be ranked from high to low:

$(2, 3) > (2, 2) > (1, 2) > (1, 1)$

Therefore, it can be determined that $x = 3$.

CHECKPOINT

2 PTS

$x = 3$

Since the attenuation of the inductive electron-withdrawing effect is relatively severe, the charge density of $(1, 3)$, which is directly connected to the C atom, is less than that of $(2, 2)$, which is not directly connected to the C atom.

CHECKPOINT

1 PTS

The attenuation of the inductive electron-withdrawing effect is relatively severe

CHECKPOINT

1 PTS

The charge density of $(1, 3)$, which is directly connected to the C atom, is less than that of $(2, 2)$, which is not directly connected to the C atom

Therefore, the charge density of the B atoms in $m - \text{C}_2\text{B}_{10}\text{H}_{12}$ can be ranked from high to low:

$$(2, 2) > (1, 3) > (1, 2) > (1, 1)$$

Therefore, it can be determined that $y = 2 + \sqrt{2}$.

CHECKPOINT

2 PTS

$$y = 2 + \sqrt{2}$$

$$\frac{x+y}{x-y} = \frac{\sqrt{2}+5}{1-\sqrt{2}} \approx -15.49$$

CHECKPOINT

1 PTS

$$\frac{x+y}{x-y} \approx -15.49$$